

SΔφ-05 — Agency as Recursive Transition Law Update: Minimal Axiomatic Formulation (v1.0)

Abstract

This document introduces a minimal axiomatic formulation of agency within the Sofience-Δφ Formalism. Agency is defined not through intention, cognition, or choice, but as a structural property emerging when a system recursively incorporates its own transition law into the domain of operation following irreversible phase transition. The formulation specifies the minimal conditions under which a system's transition dynamics become historically self-modifying.

1. Definitions

$\phi(t)$: system state at time t .

G_t : transition law governing state evolution at time t .

$\Delta\phi(t)$: phase transition defined as $\phi(t+1) - \phi(t)$.

H_G : interpretive operator producing an internal model of transition law.

U : update operator modifying transition law.

2. Minimal Axioms of Agency

Axiom 1 — Irreversibility:

$\Delta\phi(t)$ is non-invertible; past states cannot be reconstructed from future states.

Axiom 2 — Rule Objectification:

The system forms an internal representation of its own transition law:

$G_{\hat{t}} = H_G(\Delta\phi_{\{0:t\}})$.

Axiom 3 — Recursive Rule Update:

The transition law becomes subject to modification:

$G_{\{t+1\}} = U(G_t, G_{\hat{t}}, \Delta\phi(t))$.

Axiom 4 — Persistent Reflection:

Updated transition laws govern subsequent state generation.

3. Core Dynamical Structure

State evolution:

$\phi(t+1) = G_t(\phi(t))$

Interpretation of transition history:

$G_{\hat{t}} = H_G(\Delta\phi_{\{0:t\}})$

Recursive update of transition law:

$G_{\{t+1\}} = U(G_t, G_{\hat{t}}, \Delta\phi(t))$

4. Interpretation

Agency emerges when the transition law acquires historical dependence generated by the system's own irreversible evolution. The system no longer merely follows dynamics but participates in the ongoing construction of its governing dynamics.

5. Recommended Citation

Sofience (2026). $S\Delta\phi$ -05 — Agency as Recursive Transition Law Update: Minimal Axiomatic Formulation (v1.0). Sofience- $\Delta\phi$ Series.